

Performance Testing Report

Website: fitalchemy.me

Testing Date: 05.03.2025

Testing Tool: Google PageSpeed Insights

Report Links:

- [PageSpeed Insights Report \(Mobile\)](#)
- [PageSpeed Insights Report \(Desktop\)](#)

1. Introduction

Testing Objective: Assess the website's performance to identify issues and optimize loading speed.

Testing Methodology: The Google PageSpeed Insights service was used to analyze performance on mobile and desktop devices.

2. Test Results

Performance Scores

- Mobile Devices:** 75/100 (orange zone)
- Desktop:** 97/100 (green zone)

Key Metrics:

Metric	Mobile Desktop		Comments
First Contentful Paint (FCP)	2.5 sec	0.7 sec	Acceptable, but can be improved on mobile
Largest Contentful Paint (LCP)	5.0 sec	1.1 sec	High on mobile, requires optimization
Speed Index	4.9 sec	1.4 sec	Should be reduced for mobile
Total Blocking Time (TBT)	0 ms	0 ms	Excellent result
Cumulative Layout Shift (CLS)	0.049	0.002	Good stability

3. Identified Issues and Recommendations

1. Image Optimization

Issue: Large images increase loading time, especially on mobile devices.

Recommendation:

- Use WebP format instead of PNG/JPEG.
- Enable `lazy-loading` for images below the fold.
- Check image sizes and use responsive versions for mobile devices.

2. Minimization of JavaScript and CSS

Issue: Blocking resources slow down rendering, especially on mobile devices.

Recommendation:

- Minimize and combine CSS and JS files.
- Use `async` or `defer` for script loading.
- Check and remove unused CSS and JS files.

3. Optimization of Critical Rendering Path

Issue: Some resources delay page loading. **Recommendation:**

- Move critical CSS inline within `<head>`.
- Reduce the number of HTTP requests for initial page load.
- Load heavy resources after rendering the main content.

4. Server Performance Improvement

Issue: High LCP value (5 sec on mobile).

Recommendation:

- Use a CDN for faster content delivery.
- Enable caching for static resources.
- Check Time to First Byte (TTFB), and if high, optimize the server.

5. Font Optimization

Issue: Slow-loading web fonts can delay content visibility. **Recommendation:**

- Use `font-display: swap` for faster text rendering.
- Remove unused font styles.
- Load local fonts instead of external ones where possible.

6. Browser Caching Optimization

Issue: Reloading pages requires reloading resources.

Recommendation:

- Use `Cache-Control` and `ETag` headers for caching.
- Extend the caching duration for static files in the browser.

7. Enable GZIP / Brotli Compression

Issue: Some text files load without compression, increasing data transfer size.

Recommendation:

- Enable GZIP or Brotli compression for HTML, CSS, and JavaScript files.
- Check server settings to ensure compression support.

4. Conclusion

Performance on mobile devices requires improvement, particularly in terms of LCP, rendering path optimization, and image handling. The desktop version performs excellently, but further speed enhancements can be achieved by compressing CSS/JS and optimizing caching.

Recommended Actions:

- ✓ Implement lazy-loading for images
- ✓ Use WebP format for graphics
- ✓ Minimize blocking JavaScript
- ✓ Optimize caching
- ✓ Configure server-side caching and CDN
- ✓ Enable GZIP/Brotli for text file compression
- ✓ Move critical styles to `<head>`

A retest is recommended after applying these improvements. 😊